Submission Summary

Conference Name
International Conference on Sustainable Smart Computing and Communications

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Paper Title
Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

Abstract
Intrusion Detection Systems (IDS) are essential for protecting IoT networks; however, many deep learning–based IDS solutions are too computationally intensive to be deployed on resource-constrained edge devices. This paper proposes TSANet , a lightweight Temporal–Spatial Attention Network for real-time intrusion detection on low-power hardware such as the Raspberry Pi 5. The proposed model integrates a 1D Convolutional Neural Network for spatial feature learning, an LSTM network for temporal sequence modeling, and attention mechanisms to highlight important traffic features and time steps. TSANet is trained and evaluated on the CICIDS2017 dataset and optimized using pruning and quantization-aware training to reduce model size and inference latency. Experimental results show that the optimized model achieves 98.3% offline accuracy and 96.8% real-time detection accuracy on Raspberry Pi, with an inference latency of 8–12 ms per flow, demonstrating its effectiveness for practical IoT edge based intrusion detection.

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Submission Files
Lightweight_Real-Time_Intrusion_Detection_for_IoT_Networks_Using_TSANet_Deployed_on_Raspberry_Pi_with_Live_Traffic_Monitoring.pdf (822.8 Kb, 12/27/2025, 11:50:46 AM)

Submission Questions Response

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